ANKUSH MALOTHIA / SIROSIU

AI SYSTEMS | SOFTWARE ENGINEERING | DIGITAL DESIGN & RF
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PROFILE

ECE undergraduate at NIT Jalandhar building projects across applied ML, agentic AI, full-stack systems, and antenna simulation. Hands-on with Python/TypeScript, API-driven applications, local-LLM experimentation, and computational electromagnetics using CST Studio Suite and openEMS. Interested in efficient AI inference, computer architecture, and hardware-software integration.

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar
B.Tech, Electronics and Communication Engineering | Expected 2028

SELECTED PROJECTS

Ave - Personal AI Agent & Wearable OS | ONGOING PROTOTYPE

Agent architecture, local LLM experimentation, tool use, memory, voice interaction

- Building a privacy-first personal agent for natural-language interaction, tool execution, and personalized memory.
- Prototyping voice-and-text interaction while exploring quantized local inference, wearable integration, and controlled device actions.

FraudLens - Credit Card Fraud Monitoring | FULL-STACK PROTOTYPE REPO

Python, XGBoost, scikit-learn, FastAPI, WebSockets, SQLite, Next.js, TypeScript

- Trained an XGBoost transaction classifier on Kaggle's Credit Card Fraud Detection dataset; used class weighting for severe class imbalance.
- Integrated FastAPI scoring and alert endpoints with WebSocket updates and SQLite persistence; built a Next.js dashboard for review and approve/block/escalate decisions.

SIROSIU - Personal Website & Creative Engineering | ONGOING REPO

React, TypeScript, Vite, Three.js, React Three Fiber, Cloudflare Pages

- Building a 3D-enhanced portfolio that combines engineering, visual design, and interaction; live site: sirosiu.com.

Antenna Modeling & EM Simulation | LAB / SIMULATION

CST Studio Suite, openEMS (FDTD), MATLAB, ParaView

- Simulated a 2.4 GHz half-wave dipole in CST, observing resonance near 2.34 GHz and minimum VSWR of 1.01; also modeled an inset-fed patch and a broadside array near 2.6 GHz.
- Used openEMS and MATLAB to model a 5.8 GHz patch and helical antennas at 1 and 2.4 GHz; analyzed S-parameters, impedance, and near-field-to-far-field patterns, exporting 3D results to ParaView.

TECHNICAL SKILLS

Programming	Python, JavaScript, TypeScript, C++ (developing)
AI / ML	XGBoost, scikit-learn, pandas, imbalanced classification, precision-recall analysis, Ollama, quantized local LLM exploration
Backend & web	FastAPI, REST APIs, WebSockets, SQLite, React, Next.js, Vite, Tailwind CSS, Three.js, React Three Fiber
UI/UX & product	Responsive interfaces, reusable components, interaction design, 3D web experiences
Digital design & RF	Verilog (developing), Vivado, Verilator, Yosys, GTKWave; CST Studio Suite, openEMS, S-parameters, VSWR, NF2FF analysis
Tools & workflow	Linux (Arch, Ubuntu), Git/GitHub, Docker, VS Code, MATLAB, ParaView, Supabase CLI